# Pentana Vehicle Sales

## VIN Enrichment — Construction & Assessment

Synogize Internal Technical Report

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# Contents

# 1 Executive Summary

This report documents the VIN enrichment work completed for the `new_source` pipeline in the Pentana vehicle sales project. The objective was to improve data quality and field coverage by extracting trustworthy vehicle attributes from the Vehicle Identification Number (VIN), and to promote only those enrichments that are both auditable and operationally safe.

The work was implemented in two controlled releases:

- **Chunk A:** WMI-based enrichment — canonical make, manufacturer, and country of manufacture derived from the first three VIN characters.
- **Chunk B:** WMI+VDS descriptor enrichment — high-confidence null-fill candidates for seats, body style, transmission type, and fuel type.

Key outcomes at a glance:

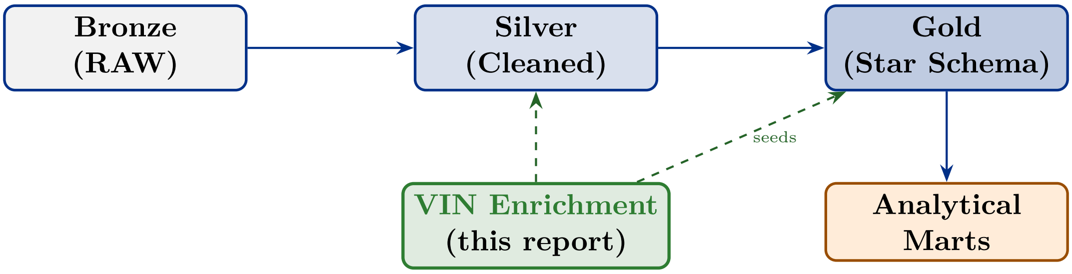
| Metric | Value | Notes |
|---|---:|---|
| Valid VINs available | 5,075,191 | strong quality base for enrichment |
| WMI dictionary rows | 797 | observed WMI universe in Silver |
| Approved WMIs | 39 + 3 | 39 fully approved; 3 with caveat |
| VINs with curated make | 4,188,938 | 82.6% of valid VINs |
| Make matches (VIN vs warehouse) | 4,151,459 | 99.1% alignment within curated set |
| Make replace candidates | 24,899 | high-confidence improvement opportunities |
| Seats candidate fills | 2,374,721 | strongest descriptor opportunity |
| Fuel type candidate fills | 784,018 | promote as null-fill first |
| Body style candidate fills | 176,433 | controlled fill candidate |
| Transmission candidate fills | 30,395 | smaller but usable |

The strongest production result was the VIN-based standardization of `MAKE`, `MANUFACTURER`, and `COUNTRY` across 4.19 million vehicle records. Descriptor enrichments derived from `WMI+VDS` were kept as controlled candidate fields rather than blind overwrites. VIN-based model year was tested but not promoted due to a 64% mismatch rate against current warehouse values.

# 2 Dataset Context

## 2.1 Position in the medallion architecture

The active workstream is `new_source`, following the standard medallion flow. VIN enrichment sits in the Silver layer and seeds both the Silver enrichment tables and the downstream Gold vehicle dimension.




## 2.2 Core source table

The VIN logic was built on top of the Silver core table:

`PENTANA_ANALYTICS.SILVER_NEW.VEHICLE_SALES_SILVER_CORE_T`

This was a deliberate design choice. Silver already contained cleaned VIN values, issue buckets, and normalized vehicle attributes needed for comparison and validation. Starting from Silver rather than Bronze avoided operating on uncleaned strings and allowed direct comparison against the attributes the VIN enrichment was intended to improve.

## 2.3 Snowflake objects created

The VIN workstream created and populated the following permanent objects in `PENTANA_ANALYTICS.SILVER_NEW`:

| Table | Purpose |
|---|---|
| `VIN_WMI_DICTIONARY_T` | Curated WMI-to-make/manufacturer/country mapping |
| `VIN_WMI_VDS_DICTIONARY_T` | Descriptor candidates keyed by WMI+VDS (8 chars) |
| `VEHICLE_VIN_ENRICHMENT_SILVER_T` | Per-VIN enrichment output and candidate flags |

# 3 VIN Structure and Decoding Approach

## 3.1 VIN anatomy

A standard 17-character VIN encodes manufacturer identity and vehicle descriptor information in well-defined positional zones. The enrichment work exploited two of these zones:

| WMI<br>pos. 1–3 | VDS<br>pos. 4–8 | VIS<br>pos. 9–17 |
|---|---|---|
| Make, Manufacturer, Country **(Chunk A)** | Body Style, Transmission, Fuel Type, Seats **(Chunk B)** | Model Year (pos. 10), Serial Number — *research only* |

- **WMI (positions 1–3)** — World Manufacturer Identifier. Assigned by an official international registry to each manufacturer and assembly plant. Because the registry is standardized and publicly cross-referenceable (e.g. NHTSA vPIC), WMI is highly reliable for deriving `MAKE`, `MANUFACTURER`, and `COUNTRY`.
- **VDS (positions 4–8)** — Vehicle Descriptor Section. Encodes model, body style, restraint system, and powertrain. Crucially, the encoding is *manufacturer-specific*: the same character sequence means different things for different brands. Chunk B validates VDS candidates against observed warehouse distributions rather than relying on external decoding.
- **VIS (positions 9–17)** — Vehicle Identifier Section. Contains the model year character at position 10 and the production serial number. Model year was tested in this workstream but was not promoted (see Section 5.5).

## 3.2 Confidence tier thresholds

Both dictionary tables apply tiered confidence scoring before any candidate flag is set:

| Tier | Condition | Enrichment use |
|---|---|---|
| `HIGH` | ≥98% dominant-value share *and* ≥100 observed rows | Candidate fill: eligible |
| `MEDIUM` | ≥95% share *and* ≥50 rows | Used for WMI make; not for VDS descriptors |
| `LOW` | Dominant value present but below threshold | Dictionary audit only; not promoted |

Only `HIGH`-confidence mappings in `APPROVED` or `APPROVED_WITH_CAVEAT` review status are used for descriptor candidate flags. `MEDIUM` confidence is accepted for WMI make enrichment (backed by external references), but not for VDS descriptor fills where the encoding is manufacturer-dependent. `LOW`-confidence and `REVIEW_REQUIRED` rows remain in the dictionary for audit and future review cycles.

# 4 Methodology

## 4.1 Chunk A — WMI enrichment

The first release built a curated WMI dictionary by seeding 42 manually reviewed WMI records (39 fully approved; 3 approved with caveat) against an observed WMI profile derived from the Silver data. This produced a dictionary of 797 rows covering the full observed WMI universe, with review status and confidence tier for each.

The following fields were derived from the WMI dictionary and stored in `VEHICLE_VIN_ENRICHMENT_SILVER_T`:

| Output field | Description |
|---|---|
| `VIN_WMI` | Extracted 3-char WMI from the VIN |
| `VIN_MAKE_STD` | Canonical make name from the approved dictionary |
| `VIN_MANUFACTURER_STD` | Legal manufacturer name |
| `VIN_COUNTRY_STD` | Country of manufacture |
| `VIN_MAKE_CONFIDENCE` | Confidence tier of the WMI mapping |
| `VIN_MAKE_MATCH_FLAG` | `Y/N`: VIN-derived make matches current Silver make |
| `VIN_MAKE_REPLACE_CANDIDATE_FLAG` | `Y`: approved high-confidence improvement available |
| `MAKE_FINAL` | Resolved make (VIN dictionary preferred; Silver fallback) |
| `MAKE_FINAL_SOURCE` | Provenance: `VIN_WMI_DICTIONARY` or `SILVER_MAKE_CLEAN` |

## 4.2 Chunk B — WMI+VDS descriptor enrichment

The second release extended the logic to `WMI+VDS` (the first 8 VIN characters), using the observed distribution of each descriptor field across VINs sharing the same prefix. For each unique WMI+VDS combination, the dominant observed value and its confidence share were computed for four fields. Each field gets a corresponding candidate flag in `VEHICLE_VIN_ENRICHMENT_SILVER_T`, set to `Y` only when (a) the dictionary confidence is `HIGH`, (b) the VIN-derived value is non-null, and (c) the current warehouse value is already null. No existing non-null warehouse values are overwritten.

| Descriptor field | Candidate flag column |
|---|---|
| `BODY_STYLE` | `BODY_STYLE_VIN_FILL_CANDIDATE_FLAG` |
| `TRANSMISSION_TYPE` | `TRANSMISSION_VIN_FILL_CANDIDATE_FLAG` |
| `FUEL_TYPE` | `FUEL_VIN_FILL_CANDIDATE_FLAG` |
| `SEATS` | `SEATS_VIN_FILL_CANDIDATE_FLAG` |

## 4.3 Validation philosophy

The workstream followed three principles:

1. **Prefer deterministic logic.** VIN structure is standardized. Exploit what is known from the official registry rather than infer what is uncertain.
2. **Validate against the warehouse.** All enrichment outputs were compared against the existing Silver values before any promotion decision.
3. **Promote only high-confidence fields.** Enrichment was staged: first a candidate flag, then a separate explicit promotion decision after human review. This prevents an incorrect enrichment from silently overwriting production data.

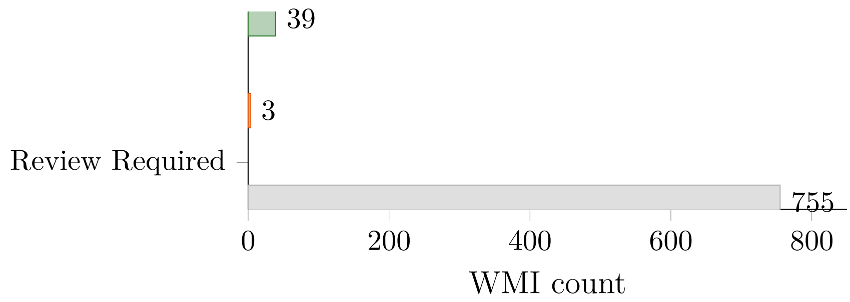
# 5 Main Findings

## 5.1 VIN quality and coverage

VIN coverage was strong enough to support meaningful enrichment. About 5.075 million rows carried valid VINs, and the quality profile was stable enough to support both WMI-level identity resolution and VDS-level descriptor research.

## 5.2 WMI dictionary coverage

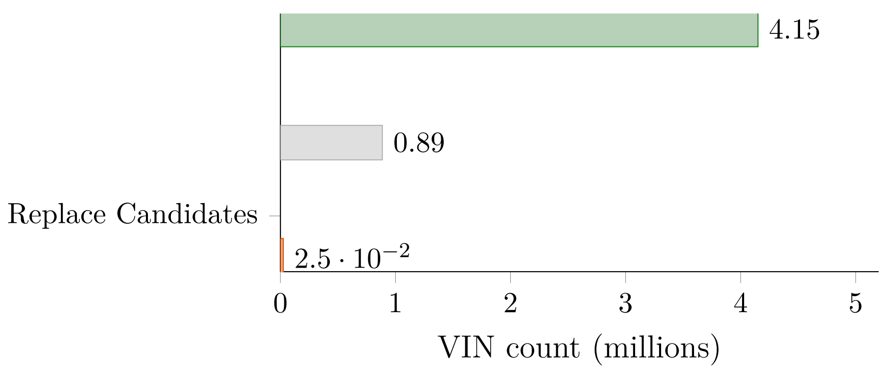
Of the 797 observed WMIs in Silver, 42 have been reviewed and approved. The remaining 755 are profiled in the dictionary at `REVIEW_REQUIRED` status — they are available for future review cycles but are not used in any enrichment output.




Expanding the reviewed set from 42 to the top 100 WMIs by observed VIN count would likely raise approved coverage above 95% of the valid VIN base. The framework is already in place — this is a data review task, not a code change.

## 5.3 VIN make enrichment outcome

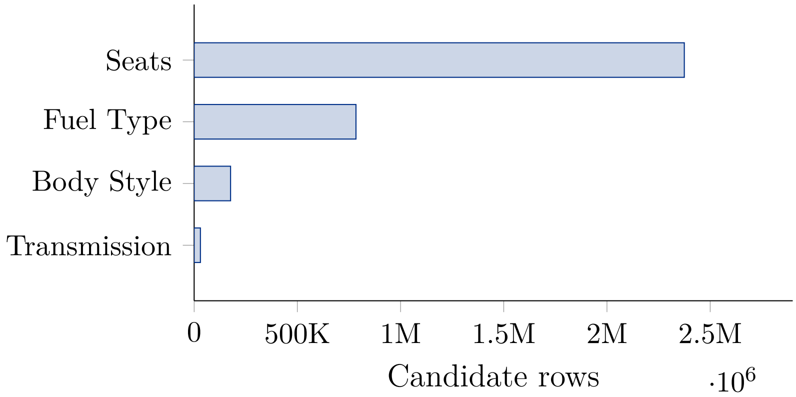
Within the 4.19 million VINs where an approved WMI mapping exists, the VIN-derived make aligned with the current Silver value for 99.1% of records. The remaining 1% represent genuine improvement candidates (label normalization differences, short codes, or null makes that can now be filled).




Most mismatches were label normalization differences (e.g. short codes vs. full names, or slightly different capitalizations from the DMS), not genuine data conflicts. The curated `VIN_MAKE_STD` resolves these systematically.

## 5.4 Descriptor fill opportunities

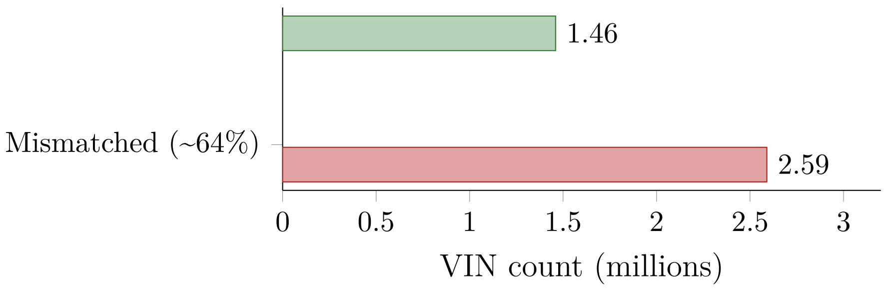
The chart below shows the number of null-fill candidate rows per descriptor field — rows where the VIN dictionary holds a `HIGH`-confidence value and the current warehouse value is null:




Seats is by far the largest single enrichment opportunity and represents a straightforward null-fill path with no overwrite risk. Fuel type has the most variance across WMI+VDS combinations and should be promoted as a null-fill only operation initially.

## 5.5 Model year outcome

VIN position 10 encodes model year using an ISO-standardized character map. The extracted model year was compared against the current `MODEL_YEAR` in Silver. The result was a 64% mismatch rate:




A 64% mismatch rate is too high to trust as a production replacement source. The three main contributing factors are:

- **DMS vs. build year.** The DMS tends to record the model year as perceived at time of sale, which may differ from the build year encoded in the VIN (common for vehicles sold late in a calendar year with the following model year designation).
- **Carry-over models.** Some manufacturers continue a model year designation into the next calendar year, so the VIN year and the commercially marketed year differ by design.
- **Pre-delivery and pre-compliance VINs.** Vehicles that cross a calendar year boundary between manufacture and registration can straddle two model year cycles.

Model year was therefore left as a research path only and is not promoted into the operational model.

# 6 Current Output Snapshot

## 6.1 WMI dictionary coverage

| Metric | Value | Notes |
|---|---|---|
| WMI dictionary rows | 797 | full observed WMI universe |
| Approved WMIs | 39 | high-confidence reviewed mappings |
| Approved with caveat | 3 | usable, noted exceptions |
| Review required | 755 | profiled; not used in enrichment |
| VINs covered by approved/caveated WMIs | 4,164,903 | 82.1% of valid VINs |

## 6.2 Vehicle enrichment output

| Metric | Value | Interpretation |
|---|---|---|
| Total valid VINs | 5,075,191 | enrichment base |
| VINs with curated make | 4,188,938 | approved WMI dictionary hit |
| Make matches | 4,151,459 | 99.1% alignment within curated set |
| Make replace candidates | 24,899 | high-confidence make improvements |
| Body style fill candidates | 176,433 | HIGH-confidence null-fill opportunity |
| Transmission fill candidates | 30,395 | HIGH-confidence null-fill opportunity |
| Fuel type fill candidates | 784,018 | promote as null-fill first |
| Seats fill candidates | 2,374,721 | largest single enrichment opportunity |

# 7 Promotion Decision

## 7.1 Status summary

| Field | Status | Rationale |
|---|---|---|
| MAKE | Promoted | WMI is an official standard; 99.1% warehouse alignment; remaining mismatches are label normalization only |
| MANUFACTURER | Promoted | Derived directly from the WMI dictionary; same reliability as make |
| COUNTRY | Promoted | WMI encodes country of manufacture; reliable for all approved mappings |
| SEATS | Candidate | Largest fill opportunity (2.37M rows); HIGH-confidence null-fill only |
| BODY_STYLE | Candidate | 176K fill candidates; controlled promotion recommended |
| TRANSMISSION_TYPE | Candidate | 30K fill candidates; smaller group but high confidence |
| FUEL_TYPE | Candidate | 784K candidates; more variance across WMI+VDS keys; null-fill first |
| MODEL_YEAR | Not promoted | 64% mismatch vs. current Silver; DMS/build-year misalignment is structural |
| External decoder | Not promoted | NHTSA vPIC and Australian RAV reserved for a later enrichment phase |

## 7.2 Recommended promotion order

For the next production release, the following order is recommended:

1. MAKE — already operationalized in Chunk A; propagate to DIM_VEHICLE and downstream Gold marts.
2. SEATS — largest gain; clean null-fill pattern with no overwrite risk.
3. BODY_STYLE — controlled fill; cross-check against existing non-null values to validate coverage before promoting.
4. TRANSMISSION_TYPE — smaller candidate group; straightforward null-fill.
5. FUEL_TYPE — null-fill first pass; review HIGH-confidence non-null replacements in a separate step before applying broadly.

# 8 Business Value

The VIN enrichment workstream delivered value at three levels.

### 1. Canonical make standardization

The curated WMI dictionary resolves label inconsistencies that originate in the DMS: short codes, variant spellings, and legacy names that prevent reliable joins between the sales fact table and the dealer dimension. A single canonical `MAKE` value, backed by a standardized external reference (NHTSA WMI registry), is a prerequisite for accurate brand-level reporting, dealer performance aggregation, and the Gold `DIM_VEHICLE` dimension.

### 2. Second-wave descriptor fills

Seats, body style, transmission, and fuel type are null for a non-trivial proportion of records. VIN-derived fills are deterministic — they depend on the vehicle's own identifier rather than statistical imputation — and are therefore safer and more auditable. At 2.37 million candidate rows, seats alone represents the largest single data quality improvement available without an external data feed.

### 3. A reusable enrichment framework

By building a dictionary-based approach with tiered confidence scoring, explicit review status, and candidate-flag outputs rather than direct overwrites, the workstream established a pattern that can be extended to future external enrichment sources (NHTSA vPIC, Australian RAV, PPSR) without requiring a redesign of the enrichment layer. The separation between *candidate* and *promoted* is a safety mechanism, not a limitation.

At the current promotion scope, the VIN work adds a canonical make layer to **4.19 million** vehicle records and creates a high-confidence fill path for up to **2.37 million** additional seat-count values.

# 9 Implementation Footprint

| File | Purpose |
| --- | --- |
| sql/new_source/16_vin_enrichment_chunk_a.sql | WMI dictionary and enrichment table (Chunk A) |
| sql/new_source/17_vin_enrichment_chunk_b.sql | WMI+VDS dictionary and descriptor candidates (Chunk B) |
| scripts/new_source/16_vin_enrichment_chunk_a.py | Python runner for Chunk A |
| scripts/new_source/17_vin_enrichment_chunk_b.py | Python runner for Chunk B |
| notebooks/new_source/15_enrichment_quality_workbench.ipynb | Quality validation workbench |

The summary note for this workstream is at:

`docs/new_source/16_vin_enrichment_summary.md`

All files are under `internship_pentana_eda/`.

# 10 Recommended Next Steps

## 10.1 Promote Chunk A make enrichment

Apply `MAKE_FINAL` from `VEHICLE_VIN_ENRICHMENT_SILVER_T` to the Silver vehicle dimension and propagate to the Gold `DIM_VEHICLE` table. The 24,899 replace candidates should be reviewed against the dealer report proxy key logic before broad promotion to avoid downstream key mismatches on `DEALER_PROXY_KEY`, which is built from `DEALERPOSTCODE_CLEAN|MAKE`.

## 10.2 Stage Chunk B descriptor fills

- Promote `SEATS` first — largest gain, clean null-fill pattern.
- Follow with `BODY_STYLE` and `TRANSMISSION_TYPE`.
- Promote `FUEL_TYPE` as a null-fill only pass; review HIGH-confidence non-null replacements separately before applying.

## 10.3 Expand WMI review coverage

755 WMIs are profiled but remain at `REVIEW_REQUIRED` status. Extending the reviewed set from 42 to the top 100 WMIs by observed VIN count would likely cover >95% of the valid VIN base. The framework is already in place — this is a data review task, not a code change.

## 10.4 External enrichment pipeline

The natural next enrichment phase, in recommended order:

1. **NHTSA vPIC** — broad decoder and spec variables (engine, displacement, GVWR, recall eligibility). Best suited to batch API calls against the curated VIN list.
2. **Australian RAV** — build, compliance, and spec fields specific to the Australian market.
3. **PPSR and recall sources** — vehicle status and history signals for financial-risk and compliance use cases.

## 10.5 Data quality monitoring

The following checks should be added to the Silver/Gold pipeline:

1. Assert that every VIN with `VIN_ISSUE_BUCKET = 'VALID'` has a matching row in `VEHICLE_VIN_ENRICHMENT_SILVER_T`.
2. Alert when new WMIs appear with `REVIEW_REQUIRED` status and observed VIN count above a threshold (e.g. 1,000 rows) — these are candidates for the next review cycle.
3. Track the fill rate of promoted descriptor fields over time as new data is ingested to confirm enrichment is being applied correctly.

# 11 Conclusion

The VIN enrichment workstream materially improved the Pentana vehicle model by delivering a trustworthy canonical make layer and a second-wave set of controlled descriptor enrichments. The work also established clear boundaries: VIN is highly effective for manufacturer identity, but not every attribute extractable from VIN is reliable enough to promote without warehouse validation.

Two lessons generalize beyond this project:

- **WMI-level enrichment is nearly always worth doing.** The identifier is standardized, the registry is public, and coverage in large automotive datasets is typically high. The main investment is the human review step for the seed table.
- **VDS-level inference requires warehouse validation.** The same VDS sequence encodes different attributes across manufacturers. Confidence thresholds and null-fill-first strategies are essential; blindly decoding VDS against a published table will introduce errors.

For the project overall, VIN enrichment delivered one of the strongest early quality gains in the `new_source` pipeline and created a reusable, auditable framework for confidence-based enrichment that can be extended to external data sources in future phases.

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This report was generated as part of the Synogize internship programme. All VIN decoding logic was built from warehouse-observed data and publicly available WMI references. Figures are accurate as of April 2026.